MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION
FIFTH SEMESTER
(2024 ADMISION ONWARDS)
Discipline Specific Elective (DSE) - MG5DSEHSC302 – HUMAN PHYSIOLOGY AND MICROBIOLOGY

Duration: 2 Hours **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Short answer type questions
Answer any **10** questions
Each question carries **2** marks

1. Define cardiac output and mention its normal value. [U] [CO1]

2. What are the major functions of haemoglobin? [U] [CO1]

3. Differentiate between systemic circulation and pulmonary circulation. [U] [CO1]

4. What is hypoxia? Mention any two causes of hypoxia. [U] [CO1]

5. What are the functions of hydrochloric acid in the stomach? [U] [CO2]

6. Define absorption and mention the major site of nutrient absorption. [U] [CO2]

7. What is a nephron? Name its major parts. [K] [CO2]

8. What is homeostasis? Mention its importance in kidney function. [U] [CO2]

9. Name any four endocrine glands and one hormone secreted by each. [K] [CO3]

10. What are the functions of insulin and glucagon? [U] [CO3]

11. Define menstruation and menopause. [U] [CO3]

12. What is Gram staining? Mention its importance in bacterial identification. [U] [CO4]

[2 × 10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions.

Each question carries ***5*** marks.

13. Explain the composition of blood and describe the functions of its major components. [U] [CO1]

14. Describe the cardiac cycle and explain its significance. [U] [CO1]

15. Explain the mechanism of respiration and the role of the respiratory system in gaseous exchange. [U] [CO1]

16. Describe the digestion and absorption of carbohydrates in the human digestive system. [U] [CO2]

17. Explain the structure and functions of the nephron and the process of urine formation. [U] [CO2]

18. Describe the major endocrine glands and their principal hormones. [U] [CO3]

19. Explain the structure and functions of the female reproductive system and the physiology of menstruation. [U] [CO3]

20. Describe the factors affecting the growth of microorganisms and explain any methods used to control microbial growth. [U] [CO4]

[5 × 6 = 30]

Part C

Essay Type Questions

Answer any ***2*** questions.

Each question carries ***10*** marks.

21. Explain the regulation of body temperature and maintenance of homeostasis in the human body. [An] [CO1]

22. Describe the physiology of the female reproductive cycle and explain the hormonal changes associated with it. [An] [CO3]

23. Explain the mechanism of action of digestive enzymes and their role in the digestion of food. [An] [CO2]

24. Describe the different methods of sterilization and disinfection used for controlling microorganisms. [An] [CO4]

[10 × 2 = 20]